

Name: _____

Date: _____

VOLUME AND SURFACE AREA, (SURFACE AREA OF CYLINDERS AND SPHERES)

LO: I can calculate the surface area of cylinders and spheres.

Surface area formulae

The **surface area** of a **cylinder** = $2\pi r^2 + 2\pi rh$ (two circles + curved surface). The surface area of a **sphere** = $4\pi r^2$.

Units: Surface area is measured in square units (cm^2 , m^2).

Key formulae

●	Cylinder SA = $2\pi r^2 + 2\pi rh$	two circles + rectangle
●	Sphere SA = $4\pi r^2$	four times pi r squared
●	Hemisphere SA = $3\pi r^2$	half sphere + circle base
●	Cylinder SA = $\pi r^2 h$	that is volume
●	Sphere SA = $2\pi r^2$	must be $4\pi r^2$

Key idea

Cylinder: unwrap to get rectangle ($2\pi rh$) plus two circles ($2\pi r^2$). Sphere: $4\pi r^2$.

Cylinder: $2\pi r(r + h)$ | **Sphere:**

$4\pi r^2$

Remember: surface area uses squared units

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – cylinder SA

Find the SA of a cylinder with $r = 3$, $h = 10$.

$$\begin{aligned} \text{SA} &= 2\pi r^2 + 2\pi rh \\ &= 2\pi(9) + 2\pi(3)(10) \\ &= 18\pi + 60\pi = 78\pi = 245.0 \text{ cm}^2 \end{aligned}$$

245.0 cm²

Add the areas of two circles and the curved surface.

Example 2 – sphere SA

Find the SA of a sphere with radius 5 cm.

$$\begin{aligned} \text{SA} &= 4\pi r^2 \\ &= 4\pi(25) \\ &= 100\pi = 314.2 \text{ cm}^2 \end{aligned}$$

314.2 cm²

Substitute into $4\pi r^2$.

Example 3 – hemisphere

Find the total SA of a hemisphere with radius 4 cm.

$$\begin{aligned} \text{Curved surface} &= 2\pi r^2 = 2\pi(16) = 32\pi \\ \text{Flat circle} &= \pi r^2 = 16\pi \\ \text{Total} &= 48\pi = 150.8 \text{ cm}^2 \end{aligned}$$

150.8 cm²

Hemisphere = half sphere surface + flat circular base.

Example 4 – open cylinder

A cylindrical can (no lid) has $r = 4$, $h = 8$. Find SA.

$$\begin{aligned} \text{Base circle: } \pi r^2 &= 16\pi \\ \text{Curved surface: } 2\pi rh &= 64\pi \\ \text{Total} &= 80\pi = 251.3 \text{ cm}^2 \end{aligned}$$

251.3 cm²

Open cylinder: one circle + curved surface.

YOUR TURN – PRACTICE (deliberate practice)

1. Cylinder surface area

Find the total SA (1 dp). Use $\pi = 3.14$.

- a) $r = 2$, $h = 5$
- b) $r = 4$, $h = 10$
- c) $r = 3$, $h = 7$
- d) $r = 6$, $h = 4$

2. Sphere surface area

Find the SA (1 dp).

- a) $r = 3$
- b) $r = 7$
- c) $r = 10$
- d) $d = 8$ ($r = 4$)

3. Hemispheres

Find the total SA including the flat base (1 dp).

- a) $r = 5$
- b) $r = 2$
- c) $r = 6$
- d) $r = 8$

4. Open/closed cylinders

Find SA for each container (1 dp).

- a) Closed cylinder: $r=5$, $h=12$
- b) Open cylinder (no lid): $r=3$, $h=8$
- c) Pipe (no ends): $r=2$, $h=20$
- d) Closed cylinder: $r=1.5$, $h=6$

5. Mixed problems

Solve each problem (1 dp).

- a) Tennis ball $r = 3.3$ cm. Surface area?
- b) Tin can $r = 4$, $h = 11$. SA for label (curved only)?
- c) Sphere SA = 200 cm^2 . Find radius.
- d) Cylinder $r=5$, SA = 440π . Find h.
- e) Earth radius 6371 km. Surface area?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Cylinder SA = $2\pi r^2 + 2\pi rh$ ☐

Correct answer: _____

Reason: _____

b) Sphere SA = $2\pi r^2$ ☐

Correct answer: _____

Reason: _____

c) The curved surface of a cylinder unwraps to a rectangle ☐

Correct answer: _____

Reason: _____

d) Surface area of a hemisphere is exactly half the sphere ☐

Correct answer: _____

Reason: _____

e) Surface area is measured in square units ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A cylinder has the same radius and height ($r = h$). If the total surface area is $150\pi \text{ cm}^2$, find the radius.

Expression: _____

Simplified: _____

b) A sphere and cylinder have the same radius r and the cylinder height equals the diameter ($h = 2r$). Show that the sphere and cylinder have the same surface area.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

